

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 30

Annexure A

Dry Run Evaluation

Code of Conduct:

I V. Madhukrishnan (192312567) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Madhukrishnan
Signature of the student:

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Dry Run.

- 1) Given the adjacency constraints forming a cycle:-
- A adjacent to B, C
 - B adjacent to A, D
 - C adjacent to A, D
 - D adjacent to B, C

Sol:-

Region	Available Colours	Assigned Colour
A	R/G/B	Red
B	R/G/B	Green
C	R/G/B	Green
D	R/G/B	Red.

Explanation:- Since A & D are not adjacent, both can be Red. B & C are not adjacent, so both can be Green.

- b) Cryptarithmic CSP:-
Problem Statement:-

$$\begin{array}{r}
 \text{S E N D} \\
 + \text{M O R E} \\
 \hline
 \text{M O N E Y}
 \end{array}$$

Each letter represents a unique digit from 0-9, and M cannot be 0

Constraints:

- All digits are unique: $S, E, N, D, M, O, R, Y \in \{0, \dots, 9\}$
- $M \neq 0$ (leading digit of Money)
- $S \neq 0$ (leading digit of SEND)
- Mathematical Constraint: $SEND + MORE = MONEY$

Converting to numerical form:-

- $SEND = 1000S + 100E + 10N + D$
- $MORE = 1000M + 1000 + 10R + E$
- $MONEY = 10000M + 10000 + 100N + 10E + Y$

Eqn:-

$$1000S + 100E + 10N + D + 1000M + 1000 + 10R + E = 10000M$$

$$+ 10000 + 100N + 10E + Y.$$

$$M = 1$$

$$\begin{array}{r} SEND \\ + 10RE \\ \hline MONEY \end{array}$$

From the hundreds column:-

- $E + 0 = N$ (or) $E + 0 + 1 = 10 + N$
- IF $E + 0 + 1 = 10 + N$
- So $E = N$ without carry from tens.

Final,

$$\begin{array}{r} 9567 \text{ (SEND)} \\ + 1085 \text{ (MORE)} \\ \hline 10652 \text{ (MONEY)} \end{array}$$

Minimax Algorithm:-

Leaf 1 = 3

Leaf 2 = 5

Leaf 3 = 2

Leaf 4 = 9

• Node 2 is a MIN node with children: Leaf 1 (3) & Leaf 2 (5)

• MIN chooses the min value among its children.

• $\text{MIN}(3, 5) = 3$

• Node 2 value = 3.

• Reasoning:- The minimizing player will choose the move that leads to value 3 because it's smaller than 5.

• Node 3 is a MIN node with children: leaf 3 (2) & leaf 4 (9)

• Min chooses the minimum value among its children

• $\text{MIN}(2, 9) = 2$

• Node 3 value = 2

• Node 1 max node with children: Node 2 (3) & Node 3 (2)

• MAX chooses the maximum value among its children

• $\text{MAX}(3, 2) = 3$

• Node 1 value = 3.

Node	children values	Value.
1	[3, 2]	3
2	[3, 5]	3
3	[2, 9]	2
Leaf 1	—	3
Leaf 2	—	5
leaf 3	—	2
leaf 4	—	9